Existing Automatic Waste-Sorting Systems and Applications

Jihyeon Kim, Savitri Gostinčar

November 2026

1 Introduction

Correct waste sorting is important for recycling, but it can sometimes be difficult to decide where an item belongs. Different countries and companies have developed smart waste bins, robotic sorting systems, and mobile applications to make this process easier.

Some systems use cameras and artificial intelligence to recognize objects. Others use sensors to detect properties such as moisture. More advanced systems can automatically move waste into separate compartments.

We researched examples from Slovenia and other countries to understand how these technologies work. We also looked at existing applications that help people sort waste using photographs, barcodes, or search functions.

2 How Automatic Waste Sorting Works

Many automatic sorting systems follow a similar process. First, the user places an item into a smart bin, or the item moves along a conveyor belt. A camera or sensor collects information about it. The system then identifies the item and decides which waste category it belongs to.

If the system includes a physical sorting mechanism, it can move the item into the correct compartment. Different systems use different mechanisms, such as movable gates or robotic arms.

Figure 1 shows a simplified example of this process.

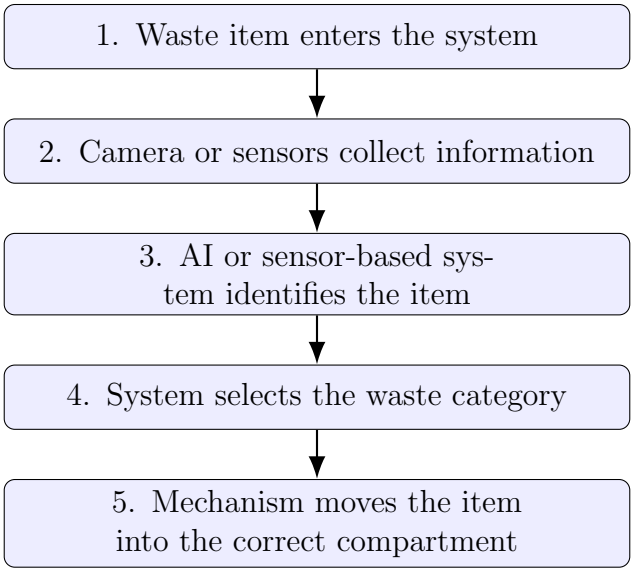


Figure 1: Simplified process of automatic waste sorting (our illustration).

3 Existing Systems in Slovenia

3.1 Trashy

Trashy is a Slovenian smart waste bin project by Dan Gantar, associated with Kersnikova.

It uses artificial intelligence and vision sensors to examine waste and help users decide how to dispose of it. The project also focuses on making waste sorting easier for visually impaired people.

This is an interesting example because the system helps people recognize waste without requiring them to know every sorting rule.

However, the available project description does not confirm that Trashy physically moves waste into separate compartments [10].

3.2 Canario

Canario is a Slovenian student project that started in September 2025. Its goal is to develop a smart bin that automatically separates wet and dry waste.

When an item is placed inside, sensors detect the object and measure its moisture. The system then uses this information to direct the item into the appropriate compartment.

The developers also describe plans to include artificial intelligence for recognizing materials such as paper and plastic.

Canario is particularly interesting for our project because it combines waste detection with physical sorting inside one bin. However, its current approach is different from our planned camera-based concept [4].

3.3 Kocerod

Kocerod is a Slovenian waste-management company that introduced an AI-powered sorting robot in cooperation with AMP Robotics.

The system uses cameras and artificial intelligence to identify waste in a sorting facility. A robotic mechanism then separates selected objects from the waste stream.

The system can also collect information about the waste being processed.

Unlike a small smart bin, Kocerod's system is designed for industrial waste sorting. It shows how artificial intelligence can be combined with robotic movement to separate waste automatically [8].

4 Existing Systems Around the World

4.1 Bin-e

Bin-e is a smart waste bin designed for shared and public spaces, such as offices, airports, and shopping areas.

The system uses artificial intelligence to recognize waste and automatically direct it into different internal compartments.

It can also compress selected waste materials, such as paper and plastic, to save space. Sensors monitor how full the compartments are, and the collected information can help with waste management.

Bin-e is one of the closest examples to our concept because it combines waste recognition, physical sorting, and several compartments in a single device [3].

4.2 TrashBot

TrashBot was developed by the American company CleanRobotics.

The system uses computer vision and artificial intelligence to recognize waste. After an item enters the bin, it is examined and classified. The system then directs it into an appropriate internal compartment.

TrashBot can be configured for different waste streams, such as recycling, general waste, and compost.

An important feature is that the system can also consider contamination. For example, food remains may affect how an item should be disposed of.

This is relevant to our project because recognizing an object does not always provide enough information for correct sorting [13].

4.3 Recycleye

Recycleye develops artificial intelligence and robotic sorting systems for recycling facilities.

In its robotic sorting systems, cameras capture images of objects moving along a conveyor belt. Artificial intelligence identifies materials and determines which objects should be removed.

A robotic arm then picks selected items and moves them into the appropriate collection area.

For example, Recycleye has described a system using a six-axis robotic arm and a pneumatic gripper. The gripper uses compressed air to hold an item before releasing it into a collection bin.

This system demonstrates how computer vision and robotic movement can work together in industrial waste sorting [5].

4.4 Research by Limsila et al.

Limsila et al. developed an automatic waste-sorting bin using computer vision.

Their system uses a camera, a Raspberry Pi, and a YOLOv5 model. The camera captures an image of an item, and the model identifies the waste type.

The recognition result is then used to control the sorting process so the item can be directed into the appropriate container.

This research is relevant because it presents a smaller automatic sorting system rather than a large industrial machine [9].

5 Comparison of Automatic Sorting Systems

System	**Recognition**	**Sorting method**
Trashy	AI and vision sensors	User guidance; physical sorting not confirmed
Canario	Moisture and object sensors	Automatic wet/dry separation
Kocerod	Cameras and AI	Industrial robotic sorting
Bin-e	AI-based recognition	Internal compartments
TrashBot	Computer vision and AI	Automatic internal sorting
Recycleye	Cameras and AI	Robotic arm or other industrial mechanisms
Limsila et al.	Camera and YOLOv5	Automatic sorting prototype

Table 1: Comparison of selected automatic waste-sorting systems.

6 Existing Waste-Sorting Applications

Besides physical waste-sorting systems, there are also applications that help people decide where to throw away different items.

These applications are especially interesting for our project because we are developing EcoSort, a web application that will help users sort waste at home.

6.1 Junker

Junker is an application developed to help people sort waste according to local recycling rules.

Users can scan the barcode of a product, search for an item by name, or take a photograph when a barcode is not available.

The application identifies the item and provides disposal instructions based on the user's area.

It also includes information about waste collection schedules and collection points.

According to its official website, Junker has a database containing more than two million products.

Junker is relevant to EcoSort because it combines item recognition with local waste-sorting information [7].

6.2 Scrapp

Scrapp is a recycling application that helps users understand how to dispose of different products.

One of its main functions is barcode scanning. The user scans a product or searches for it, and the application provides sorting instructions.

Scrapp also considers local recycling rules, which is important because waste-sorting systems can differ between cities and countries.

The application demonstrates another approach to waste identification. Instead of relying only on image recognition, it can use information connected to a product's barcode [11].

6.3 Wasteless

Wasteless is an AI-powered application designed to help users sort waste correctly.

The user points the phone camera at an item and captures an image. The application detects the object and displays sorting guidance.

According to its website, the recommendations can be adjusted to local conditions.

This application is especially similar to our EcoSort idea because the main interaction is based on taking a picture of an item and receiving a waste-sorting recommendation [14].

7 How Waste-Sorting Applications Work

Waste-sorting applications usually require the user to identify an item using a photograph, barcode, or search function.

After receiving the information, the application determines the type of item and provides instructions for disposal.

Some applications also consider the user's location because sorting rules can be different in different municipalities.

Figure 2 shows a simplified process for a photo-based application.

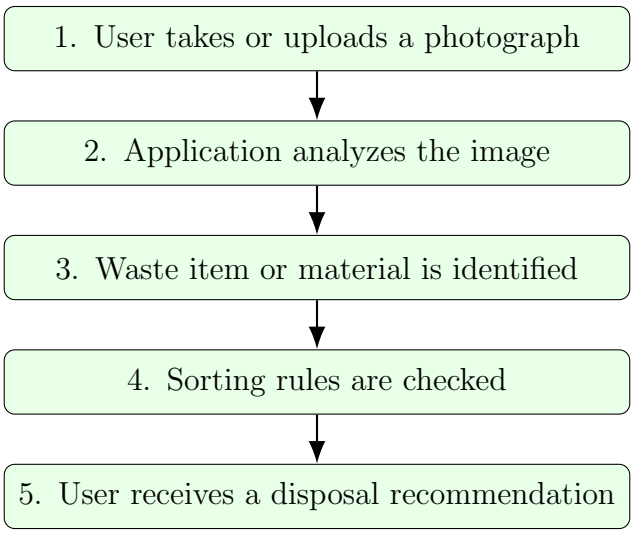


Figure 2: Simplified process of photo-based waste-sorting applications (our illustration).

8 Comparison of Waste-Sorting Applications

Application	**How users identify waste**	**Main purpose**
Junker	Barcode, photo, or search	Local sorting instructions
Scrapp	Barcode or search	Product disposal guidance
Wasteless	Camera-based recognition	AI sorting recommendations
EcoSort (planned)	Photo upload or camera	Guidance based on Ljubljana sorting rules, with follow-up questions

Table 2: Comparison of existing applications and our planned EcoSort application.

9 Challenges and Limitations

Although automatic sorting systems can be useful, they also face several challenges.

One problem is that objects can look very similar even when they are made from different materials. Transparent packaging, damaged objects, and overlapping items can also make recognition more difficult.

The ZeroWaste dataset demonstrates challenges related to recognizing deformable objects in cluttered waste environments [2].

Thielmann et al. also discuss the challenges of using computer vision in industrial waste classification [12].

Another problem is that recognizing an object does not always mean knowing how to dispose of it correctly.

For example, a pizza box may be made of cardboard, but food contamination can affect the correct disposal method. Some products also contain several materials that need to be separated.

Waste-sorting rules may also differ between municipalities. For this reason, applications should consider local guidelines instead of giving the same recommendation everywhere.

10 What We Learned for Our Project

After researching these examples, we found that smart waste bins and waste-sorting applications solve different parts of the same problem.

Automatic sorting bins can recognize waste and physically move it into the correct compartment. Applications help users understand where waste belongs when automatic sorting equipment is not available.

For our project, we would like to present a concept of a smart bin that uses computer vision and an automatic sorting mechanism.

The user would place an item into one opening. A camera would recognize it, and the system would direct it into one of five compartments.

The five categories would follow Ljubljana's VOKA SNAGA guidelines: paper and cardboard, packaging, glass, organic waste, and residual waste [6].

The smart bin will only be presented as a concept and will not be physically built.

Our practical work will focus on the EcoSort web application, developed using Base44 [1].

Users will be able to upload or take a photograph of an item and receive a waste-sorting recommendation.

If the image does not provide enough information, the application may ask additional questions, for example about the material or contamination.

We plan to test the application with approximately 30 waste items and compare its recommendations with the official Ljubljana sorting guidelines.

References

[1] Base44. Base44. `https://base44.com/`. Accessed October 2026.

[2] Dina Bashkirova, Mohamed Abdelfattah, Ziliang Zhu, James Akl, Fadi Alladkani, Ping Hu, Vitaly Ablavsky, Berk Calli, Sarah Adel Bargal, and Kate Saenko. Zerowaste dataset: Towards deformable object segmentation in cluttered scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21147–21157, 2022.

[3] Bin-e. Bin-e: Smart waste bin. `https://www.bine.world/`. Official company website. Accessed November 2026.

[4] Canario. Canario: Pametni smetnjak za cistejso zemljo. `https://canario.borstnar.com/`. Slovenian student project. Accessed November 2026.

[5] International Federation of Robotics. Fanuc robots put to use in the recycling industry. https://ifr.org/, 2022. Published March 10, 2022. Accessed November 2026.

[6] JP VOKA SNAGA. What goes where? https://vokasnaga.si/sites/www.jhl.si/files/dokumenti/what_goes_where_3r_eng_final.pdf. Accessed October 2026.

[7] Junker. Junker: Waste sorting application. https://junker.app/en/junkerinternational/. Official website. Accessed November 2026.

[8] Koroske Novice. V druzbi kocerod s pomocjo umetne inteligence nad odpadke. https://www.koroskenovice.si/novice/v-druzbi-kocerod-s-pomocjo-umetne-inteligence-nad-odpadke/, 2024. Published January 12, 2024. Accessed November 2026.

[9] Tinapat Limsila, Aphiphu Sirimangkalalo, Wasutha Chuengwutigool, and Weinian Feng. Computer-vision-powered automatic waste sorting bin: a machine learning-based solution on waste management. *Journal of Physics: Conference Series*, 2550(1):012030, 2023.

[10] Making Spaces. Trashy. https://m4kingspaces.org/project/trashy/. Project by Dan Gantar, Slovenia. Accessed November 2026.

[11] Scrapp. Scrapp recycling app. https://www.scrappzero.com/products/mobile-app. Official website. Accessed November 2026.

[12] Pelle Thielmann, Yu Zhou, Bruno Mirbach, Didier Stricker, and Jason Raphael Rambach. A review of computer vision for industrial-grade waste classification. *IEEE Access*, 13:151934–151953, 2025.

[13] U.S. Small Business Innovation Research. Cleanrobotics trashbot. https://www.sbir.gov/awards/197101. Project description. Accessed November 2026.

[14] Wasteless. Wasteless: Fighting waste with ai. https://wasteless.app/. Official website. Accessed November 2026.